

Statistical Mechanics I Exercise 5

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1 Renormalization in 1d Spin 1 System

In this section we use RG to analyse the 1d Spin 1 Hamiltonian

$$\beta H = -K \sum_{\langle i,j \rangle} S_i S_j \quad (1)$$

Where $S_i \in \{-1, 0, 1\}$. This can be rewritten as:

$$\beta H = -K \sum_i S_i S_{i+1} \quad (2)$$

Where we assumed periodic boundary conditions and set $S_{N+1} = S_1$. Now let us start by decimating this model. The partition function is:

$$Z = \sum_{S_1} \sum_{S_2} \cdots \sum_{S_N} \exp \left[-K \sum_i S_i S_{i+1} \right] \quad (3)$$

Let us now decimate by summing over the even spins. For each spin, there are two affected terms in the sum in the exponent:

$$\exp \left[-K \sum_i S_i S_{i+1} \right] = \exp \left[-K \sum_{i \text{ even}} S_i (S_{i-1} + S_{i+1}) \right] \quad (4)$$

So the partition function becomes:

$$Z = \sum_{S_1} \sum_{S_2} \cdots \sum_{S_N} \exp \left[-K \sum_{i \text{ even}} S_i (S_{i-1} + S_{i+1}) \right] = \sum_{S_1} \sum_{S_3} \cdots \sum_{S_N} \prod_{i \text{ even}} [2 \cosh K(S_{i-1} + S_{i+1}) + 1] \quad (5)$$

We can change the product to be:

$$\prod_{i \text{ even}} [2 \cosh K(S_{i-1} + S_{i+1}) + 1] = \prod_{i \text{ odd}} [2 \cosh K(S_i + S_{i+2}) + 1] = \prod_i^{N/2} [2 \cosh K(S_i + S_{i+1}) + 1] \quad (6)$$

So we can write the new Hamiltonian:

$$\beta H' = \sum_i^{N/2} \ln(2 \cosh K(S_i + S_{i+1}) + 1) \quad (7)$$

Comparing this to the previous Hamiltonian we get that they cannot fit. The term by term comparison originally includes nine equations, but reducing to the cases based on the different sum and multiplication, we get four equations, based on the three possible values for the sum up to a sign (since cosh is even), where the 0 value is split between the options of the multiplication:

$S_i S_j$	$ S_i + S_j $	(S_i, S_j) tuples
-1	0	$(-1, 1), (1, -1)$
0	0	$(0, 0)$
0	1	$(1, 0), (0, 1), (-1, 0), (-1, 0)$
1	2	$(1, 1), (-1, -1)$

This means that in the original Hamiltonian we are missing three terms, which will allow us to solve these four equations. For this, let us redefine it with the next three terms that support the current symmetries. These symmetries are "directionality", i.e. the $i \leftrightarrow j$ exchange symmetry and the $+\leftrightarrow -$ global symmetry. The Hamiltonian then becomes:

$$\beta H = \sum_i \left[-K S_i S_{i+1} - L (S_i^2 + S_{i+1}^2) - M (S_i S_{i+1})^2 - K_0 \right] \quad (8)$$

Now the decimation gives:

$$\beta H = - \sum_{i \text{ even}} K S_i (S_{i-1} + S_{i+1}) + L (S_{i-1}^2 + 2S_i^2 + S_{i+1}^2) + M S_i^2 (S_{i-1}^2 + S_{i+1}^2) + 2K_0 \quad (9)$$

And so the sum on S_i in the partition function becomes:

$$\begin{aligned} \sum_{S_i} \exp \left[- \sum_{i \text{ even}} K S_i (S_{i-1} + S_{i+1}) + L (S_{i-1}^2 + 2S_i^2 + S_{i+1}^2) + M S_i^2 (S_{i-1}^2 + S_{i+1}^2) + 2K_0 \right] = \\ = \exp \left[- \sum_{i \text{ even}} -K (S_{i-1} + S_{i+1}) + L (S_{i-1}^2 + 2 + S_{i+1}^2) + M (S_{i-1}^2 + S_{i+1}^2) + 2K_0 \right] + \\ + \exp \left[- \sum_{i \text{ even}} L (S_{i-1}^2 + S_{i+1}^2) + 2K_0 \right] + \\ + \exp \left[- \sum_{i \text{ even}} K (S_{i-1} + S_{i+1}) + L (S_{i-1}^2 + 2 + S_{i+1}^2) + M (S_{i-1}^2 + S_{i+1}^2) + 2K_0 \right] = \\ = \prod_i^{N/2} (2e^{-2L} \cosh K (S_i + S_{i+1}) + 1) e^{-(L+M)(S_i^2 + S_{i+1}^2) - 2K_0} \end{aligned} \quad (10)$$

So the decimated Hamiltonian is

$$\beta H' = \sum_i^{N/2} \ln (2e^{-2L} \cosh K (S_i + S_{i+1}) + 1) - (L + M)(S_i^2 + S_{i+1}^2) - 2K_0 \quad (11)$$

The term by term comparison is

$$\begin{aligned} -K' S_i S_{i+1} - L' (S_i^2 + S_{i+1}^2) - M' (S_i S_{i+1})^2 - K'_0 = \\ = \ln (2e^{-2L} \cosh K (S_i + S_{i+1}) + 1) - (L + M)(S_i^2 + S_{i+1}^2) - 2K_0 \end{aligned} \quad (12)$$

Which gives the equations

$$\begin{aligned} K' - 2L' - M' - K'_0 &= \ln(2e^{-2L} + 1) - 2(L + M) - 2K_0 \\ -K'_0 &= \ln(2e^{-2L} + 1) - 2K_0 \\ -L' - K'_0 &= \ln(2e^{-2L} \cosh(K) + 1) - (L + M) - 2K_0 \\ -K' - 2L' - M' - K'_0 &= \ln(2e^{-2L} \cosh(2K) + 1) - 2K_0 \end{aligned} \quad (13)$$

Organizing the equations, we get:

$$\begin{aligned}
K' &= -(L + M) - \frac{1}{2} \ln \left(\frac{2e^{-2L} \cosh(2K) + 1}{2e^{-2L} + 1} \right) \\
M' &= -(L + M) - \frac{1}{2} \ln \left(\frac{2e^{-2L} \cosh(2K) + 1}{2e^{-2L} + 1} \right) + 2 \ln \left(\frac{2e^{-2L} \cosh(K) + 1}{2e^{-2L} + 1} \right) \\
L' &= (L + M) - \ln \left(\frac{2e^{-2L} \cosh(K) + 1}{2e^{-2L} + 1} \right) \\
K'_0 &= 2K_0 - \ln(2e^{-2L} + 1)
\end{aligned} \tag{14}$$

Now let us look at the obvious fixed points. For starter, for the point $K = L = M = K_0 = 0$, all three equations are stable while the last equation causes K_0 to change by $-\ln 9$ each iteration. This is then a stable fixed point. On the other side, at $L = M = K = K_0 \rightarrow \infty$, last two equations are stable, while the first two equations are unstable. This is expected, as in 1D there are no phase transition and so the GR flow goes from $\beta = \infty$ to $\beta = 0$.

2 The central limit theorem as RG

2.1 The probability of the sum of variables

Let $p(x)$ be the probability the probability that a random variable A_1 has a value x , and let A_2 be another variable with the same probability. For every x , the probability that their sum $A_1 + A_2$ will equal that value, is the probability that A_1 will take one value y and A_2 will take the complementary value $x - y$, for every possible y . This is:

$$p(A_1 + A_2 = x) = \sum_y p(y)p(x - y) \rightarrow_{\text{continuum limit}} \int_{-\infty}^{\infty} p(x - y)p(y)dy = C[p(x)] \tag{15}$$

So, if the width of the original distribution is σ_1 , the new distribution is:

$$\begin{aligned}
\sigma_2^2 &= \langle (A_1 + A_2)^2 \rangle - \langle A_1 + A_2 \rangle^2 = \\
&= \langle A_1^2 \rangle + 2\langle A_1 A_2 \rangle + \langle A_2^2 \rangle - \langle A_1 \rangle^2 - 2\langle A_1 \rangle \langle A_2 \rangle - \langle A_2 \rangle^2 = \\
&= 2(\langle A_1 A_2 \rangle - \langle A_1 \rangle \langle A_2 \rangle) = \\
&= 2\sigma_1^2
\end{aligned} \tag{16}$$

So the new distribution has a width of $\sqrt{2}$ times the old one.

2.2 The Rescaling operator

To overcome this, let us define the rescaling operator:

$$S_{\sqrt{2}}[p(x)] = \sqrt{2}p(\sqrt{2}x) \tag{17}$$

Which follows:

$$\int_{-\infty}^{\infty} S_{\sqrt{2}}[p(x)] dx = \sqrt{2} \int_{-\infty}^{\infty} p(\sqrt{2}x) dx = \left\{ \begin{array}{l} y = \sqrt{2}x \\ dy = \sqrt{2}dx \end{array} \right\} = \int_{-\infty}^{\infty} p(y) dy = \int_{-\infty}^{\infty} p(x) dx = \tag{18}$$

So, if $p(x)$ is normalized

$$\int_{-\infty}^{\infty} p(x) dx = 1 \tag{19}$$

Then so is $S_{\sqrt{2}}[p(x)]$.

2.3 The Fourier transition of the Rescaling operator

Additionally, the Fourier transform of $S_{\sqrt{2}}[p(x)]$ follows the simple rescaling relation:

$$\begin{aligned}\mathcal{F}[S_{\sqrt{2}}[p(x)]] &= \sqrt{2}\mathcal{F}[p(\sqrt{2}x)] = \sqrt{2} \int_{-\infty}^{\infty} p(\sqrt{2}x) e^{-ikx} dx = \\ &= \left\{ \begin{array}{l} y = \sqrt{2}x \\ dy = \sqrt{2}dx \end{array} \right\} = \int_{-\infty}^{\infty} p(y) e^{-iky/\sqrt{2}} dy = \tilde{p}(k/\sqrt{2})\end{aligned}\quad (20)$$

2.4 The Coarse graining operator

Now we can define the coarse graining operator T :

$$T[p(x)] = S_{\sqrt{2}}[C[p(x)]] \quad (21)$$

This is a GR procedure, it includes the decimation over the first variable (or the second variable, since the operation is symmetric), leaving the probability similar to the original one, which rescaling similar to the length rescaling in real space GR. This gives the recursion relation:

$$p'(x) = \sqrt{2} \int_{-\infty}^{\infty} p(\sqrt{2}x - y) p(y) dy \quad (22)$$

The Gaussian then:

$$p(x) = \frac{1}{\sqrt{\pi}} e^{-x^2} \quad (23)$$

Is a fixed point:

$$\begin{aligned}T[e^{-x^2}] &= \sqrt{\frac{2}{\pi}} \int_{-\infty}^{\infty} e^{-(\sqrt{2}x-y)^2} e^{-y^2} dy = \\ &= \sqrt{\frac{2}{\pi}} \int_{-\infty}^{\infty} e^{-2x^2 + 2\sqrt{2}xy - 2y^2} dy = \\ &= \left\{ \begin{array}{l} z = \sqrt{2}y \\ dz = \sqrt{2}dy \end{array} \right\} = \\ &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-2x^2 + 2xz - z^2} dz = e^{-x^2} \int_{-\infty}^{\infty} e^{(z-x)^2} dz = p(x)\end{aligned}\quad (24)$$

2.5 Perturbation around the fixed point

Let us now take a perturbation from the fixed point $p(x) = p^*(x) + \epsilon f(x)$ where $p^*(x)$ is the fixed point. For this, we can linearize the coarse graining operator by looking at the first variation:

$$\begin{aligned}\delta T[p(x)] &= T[p(x)] - T[p^*(x)] = \sqrt{2} \int_{-\infty}^{\infty} dy [p(\sqrt{2}x - y)p(y) - p^*(\sqrt{2}x - y)p^*(y)] \approx \\ &\approx \sqrt{2} \int_{-\infty}^{\infty} dy [p^*(\sqrt{2}x - y)\epsilon f(y) + \epsilon f(\sqrt{2}x - y)p^*(y)] = \\ &= 2\sqrt{2}\epsilon \int_{-\infty}^{\infty} p^*(\sqrt{2}x - y)f(y)dy\end{aligned}\quad (25)$$

So, the first order approximation for $T[p(x)]$ is:

$$T[f(x)] \approx 2\sqrt{2} \int_{-\infty}^{\infty} p^*(\sqrt{2}x - y)f(y)dy \quad (26)$$

Let us now find the eigenvalues and eigenfunctions. For this, we will move to Fourier space:

$$\tilde{T}[f(x)] = 2\tilde{p}^* \left(\frac{k}{\sqrt{2}} \right) \tilde{f} \left(\frac{k}{\sqrt{2}} \right) \quad (27)$$

And so the eigenfunctions f_n follow:

$$\tilde{T}[f(x)] \tilde{f}_n(k) = \lambda_n \tilde{f}_n(k) \quad (28)$$

$$2\tilde{p}^* \left(\frac{k}{\sqrt{2}} \right) \tilde{f} \left(\frac{k}{\sqrt{2}} \right) = \lambda_n \tilde{f}_n(k) \quad (29)$$

$$\tilde{f}_n(k) = \frac{2}{\lambda_n} \tilde{p}^* \left(\frac{k}{\sqrt{2}} \right) \tilde{f}_n \left(\frac{k}{\sqrt{2}} \right) \quad (30)$$

Using the Gaussian polynomial:

$$p^*(x) = \frac{1}{\sqrt{\pi}} e^{-x^2} \rightarrow \tilde{p}^*(k) = e^{-k^2/4} \quad (31)$$

Plugging this in, this gives:

$$\tilde{f}_n(k) = \frac{2}{\lambda_n} e^{-k^2/8} \tilde{f}_n \left(\frac{k}{\sqrt{2}} \right) \quad (32)$$

So, $\tilde{f}_n$ are function that allow scaling up to a Gaussian. Let us factor out the Gaussian:

$$\tilde{f}_n(k) \equiv e^{-k^2/4} \tilde{g}_n(k) \quad (33)$$

$$e^{-k^2/4} \tilde{g}_n(k) = \frac{2}{\lambda_n} e^{-k^2/8} e^{-k^2/8} \tilde{g}_n(k) \left(\frac{k}{\sqrt{2}} \right) \quad (34)$$

$$\tilde{g}_n(k) = \frac{2}{\lambda_n} \tilde{g}_n \left(\frac{k}{\sqrt{2}} \right) \quad (35)$$

One Hilbert space basis that allows scaling is the polynomials:

$$\tilde{g}_n(k) = k^n \quad (36)$$

$$k^n = \frac{2}{\lambda_n} \left(\frac{k}{\sqrt{2}} \right)^n \quad (37)$$

$$k^n = \frac{2^{1-n/2}}{\lambda_n} k^n \rightarrow \lambda_n = 2^{1-n/2} \quad (38)$$

The eigenfunctions are then:

$$\tilde{f}_n(k) = e^{-k^2/4} k^n \quad (39)$$

$$f_n(x) = \mathcal{F}^{-1} \left[e^{-k^2/4} k^n \right] = (i\partial_x)^n \mathcal{F}^{-1} \left[e^{-k^2/4} \right] = \frac{1}{\sqrt{\pi}} (i\partial_x)^n e^{-x^2} = \frac{(-1)^n}{\sqrt{\pi}} H_n(x) e^{-x^2} \quad (40)$$

Where $H_n(x)$ are the Hermite polynomials:

$$H_n(x) = (-1)^n e^{x^2} (\partial_x)^n e^{-x^2} \quad (41)$$

2.6 Eigendirections

Let us look at when the eigendirections pull towards the fixed point or push away from it. For this, we look compare the eigenvalues to 1:

$$\lambda_n = 2^{1-n/2} \rightarrow \begin{cases} \lambda_n > 1 & n < 2 \\ \lambda_n = 1 & n = 2 \\ \lambda_n < 1 & n > 2 \end{cases} \quad (42)$$

Let us now look at the $\lambda_n > 1$ portion. This portion includes two eigenvalues $n = 0, 1$:

$$\begin{aligned} \lambda_0 &= 2 & f_0(x) &= \frac{1}{\sqrt{\pi}} e^{-x^2} \\ \lambda_1 &= \sqrt{2} & f_1(x) &= -\frac{2}{\sqrt{\pi}} x e^{-x^2} \end{aligned} \quad (43)$$

We can see that f_0 simply adds a constant perturbation in the same "direction" of the gaussian in the function space, and so we can think about it as "parallel" to it, it rescales the system. However, f_1 pushes in a direction perpendicular to the gaussian. Looking at the Gaussian again as a probability, this corresponds to the first moment of the distribution.

3 RG in percolation

In this section we ran a model for percolation on a 2048×2048 $2d$ square lattice. We generated each lattice with a prebuilt cite occupation probability p . The code for this section can be found [here](#).

3.1 Precolation probability as a function of p

First, we wanted to plot the probability of percolation as a function of the cite occupation probability. To do this, we started from $p = 0$ and increased in steps of 0.05. For each cite occupation probability, we generated 100 random lattices using LCG, and counted then number of lattices for which there was a precolation. To check for percolation, we used a Breadth-First Search algorithm: starting from the top row, it collects occupied cites, making sure not to go over the same cite twice, then each iteration going over a visited cite to search for other occupied cites neighboring it. It continues collecting occupied cites until one of them is at the bottom row. The graph of percolation probability as a function of cite occupation probability can be seen in figure 1. As can be seen, the precolation probability jumps sharply around $p = p_c = 0.618$, as expected for such a large lattice.

3.2 The critical cite occupation probability

Around the critical cite occupation probability $p = p_c = 0.618$, we wanted to see how the lattice changes with coarse graining. To do this, we generated one lattice at $p = p_c$, one lattice at $p = 0.55$ and one lattice at $p = 0.65$. For each of these lattices, we repeatedly applied coarse graining by a factor of 2, plotting the lattice itself and finally plotting the lattice occupation density as a function of the coarse graining iteration. The lattices can be seen in figures 4, figure 5, and 6, and the density functions can be seen in figure 2. As expected, the coarse graining operation for $p < p_c$ slowly decreases the lattice occupation density, while for $p > p_c$ it slowly increases it. for $p = p_c$, the density remains roughly the same. To see that it is actually constant, we generated four additional lattices at $p = p_c$, and plotted their average density for each coarse graining iteration. The result can be seen in figure 3. It is clear that the density at $p = p_c$ is invariant under the coarse graining operation.

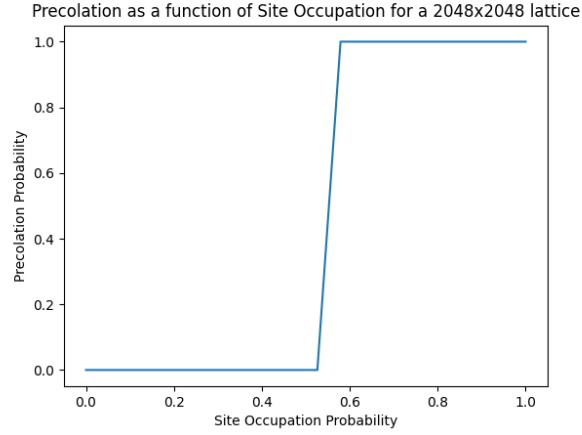


Figure 1: The graph of percolation probability as a function of cite occupation probability can be seen in figure 1. As can be seen, the precolation probability jumps sharply around $p = p_c = 0.618$, as expected for such a large lattice.

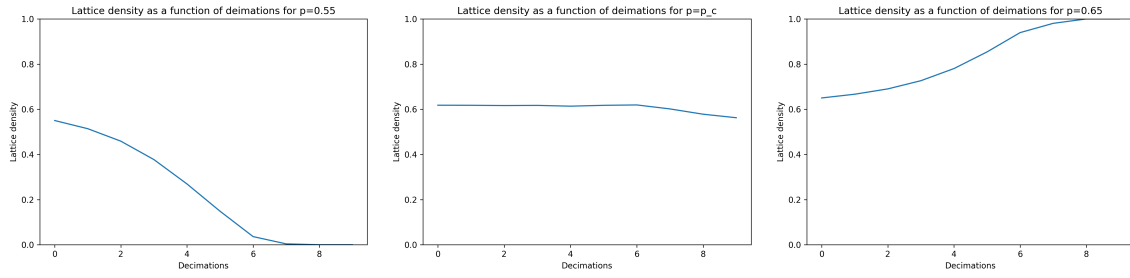


Figure 2: Lattice densities as a function of coarse graining iteration, for cite occupation probabilities $p = 0.55$ (left), $p_c = 0.618$ (center), and $p = 0.65$ (right).

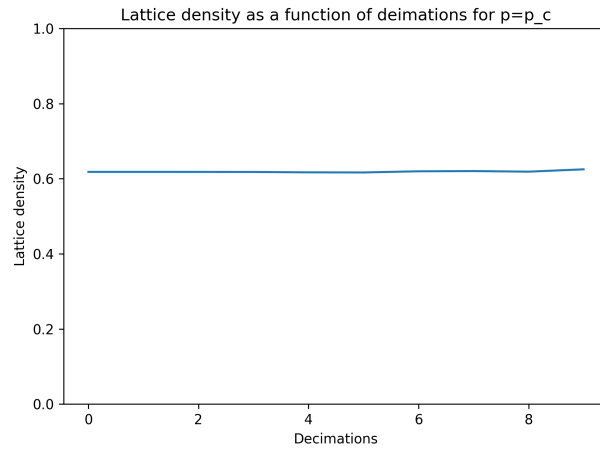


Figure 3: Average lattice densities of five randomly generated lattices as a function of coarse graining iteration, for cite occupation probabilities $p = p_c$.

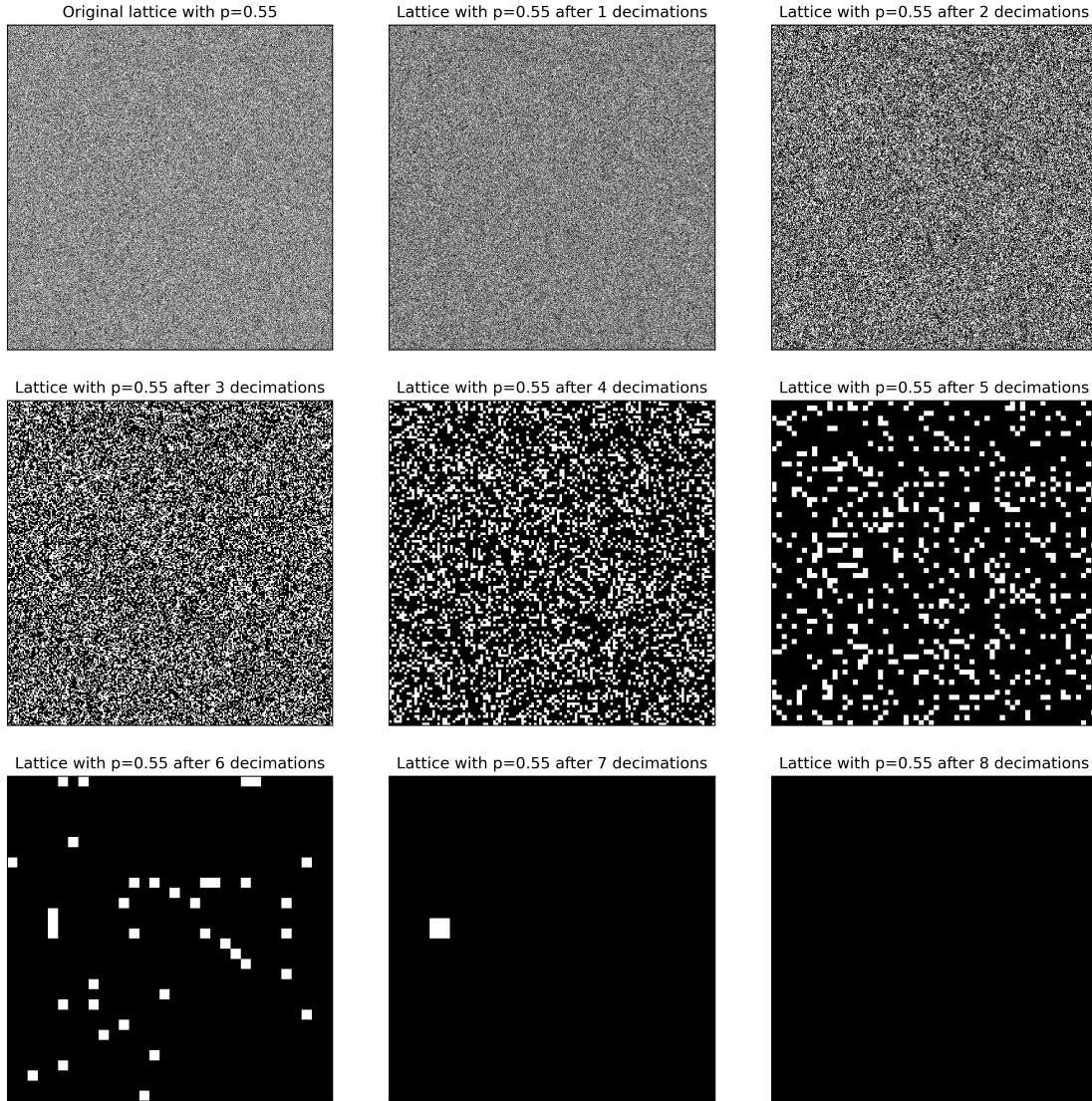


Figure 4: A lattice with $p = 0.55$ in different stages of the coarse graining. Occupied sites are white.

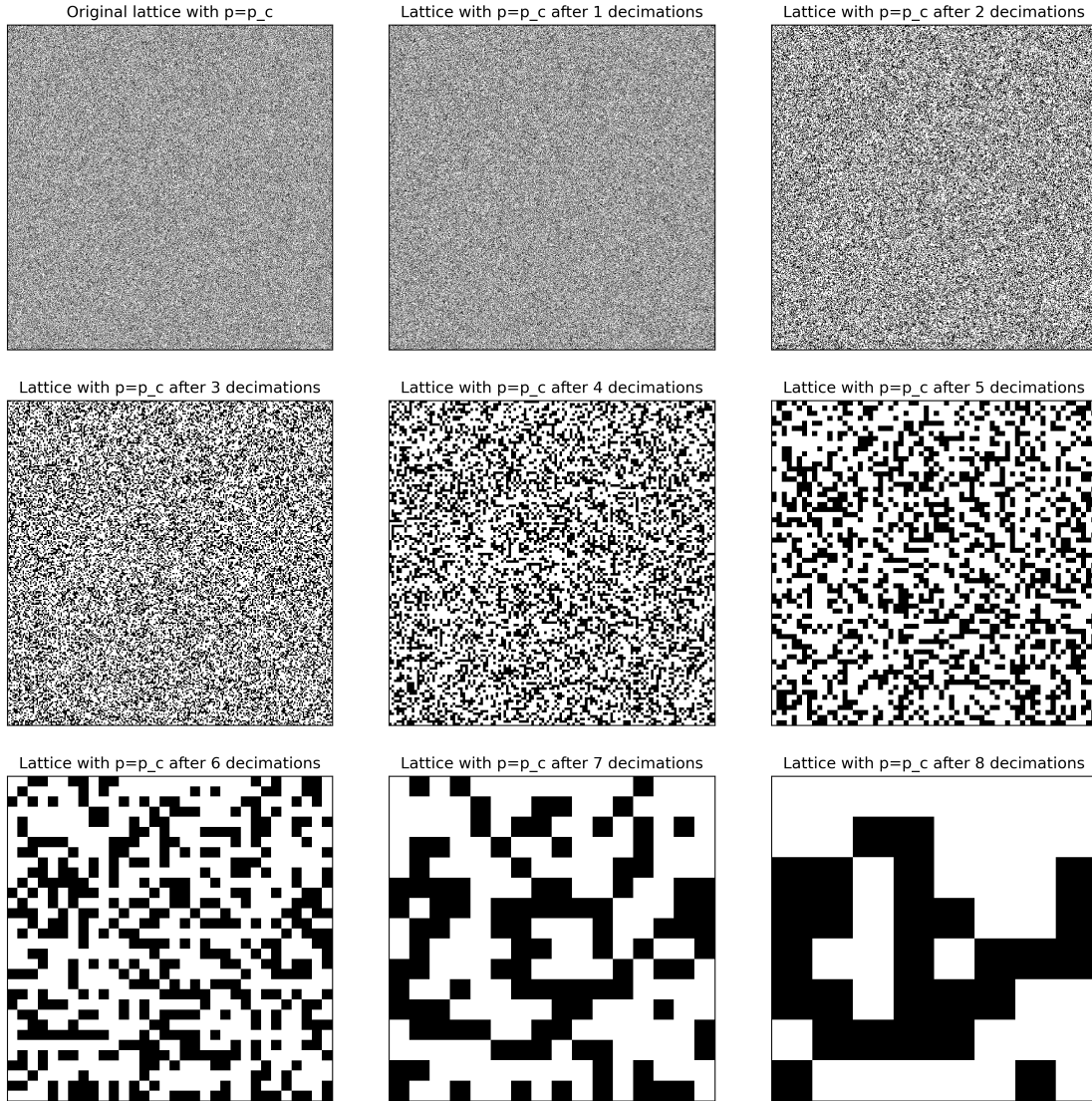


Figure 5: A lattice with $p = p_c$ in different stages of the coarse graining. Occupied sites are white.

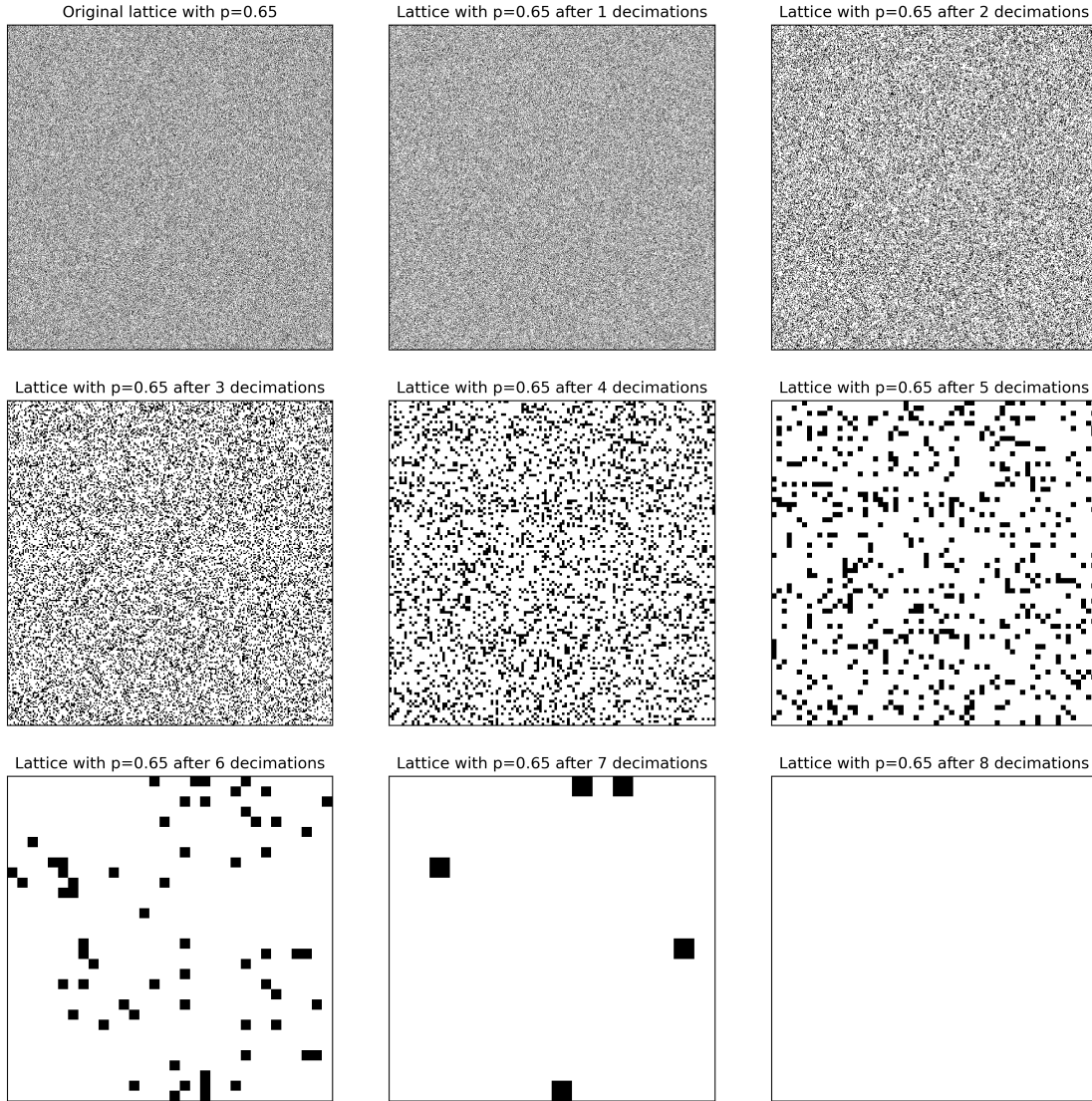


Figure 6: A lattice with $p = 0.65$ in different stages of the coarse graining. Occupied sites are white.